

AN 980: Nios[®] V Processor Quartus[®] Prime Software Support

Contents

1. Overview.....	3
2. Quartus Prime Software Support.....	4
3. Features Support and Microarchitecture.....	5
4. Board Support Package Editor.....	6
4.1. Generating the BSP in Quartus Prime Standard Edition.....	6
4.2. Generating the BSP in Quartus Prime Pro Edition.....	7
5. Document Revision History for AN 980: Nios V Processor Quartus Prime Software Support.....	9



1. Overview

The Quartus® Prime Pro Edition and Quartus Prime Standard Edition support Nios® V processor for hardware system design. Each software offers the same Nios V processor capabilities with exceptions for the following features:

- Processor and device support
- Supported features and microarchitecture
- Board Support Package (BSP) Editor generation
 - System file (QSYS, SOPCINFO)

Related Information

- [Nios® V Embedded Processor Design Handbook](#)
Describes how to most effectively use the tools, recommends design styles, and practices for developing, debugging, and optimizing embedded systems using the Nios® V processor and Altera-provided tools.
- [Nios® V Processor Reference Manual](#)
Provides information about the Nios V processor performance benchmarks, processor architecture, the programming model, and the core implementation.
- [Nios® V Processor Software Developer Handbook](#)
Describes the Nios® V processor software development environment, the tools that are available, and the process to build software to run on Nios® V processor.
- [Nios® V Processor IP Release Notes](#)
For more information about the IP version, corresponding software version, and the latest release for the Nios® V processor.

2. Quartus Prime Software Support

Table 1. Nios V Processor Core and Device Support

Nios V Processor	Quartus Prime Standard Edition Software	Quartus Prime Pro Edition Software
Processor core	- Nios V/c processor - Nios V/m processor - Nios V/g processor	- Nios V/c processor - Nios V/m processor - Nios V/g processor
Device support	- Control block-based device - MAX[®] 10 - Cyclone[®] IV E - Cyclone IV GX - Cyclone V - Cyclone 10 LP - Arria[®] 10 - Arria II GX - Arria II GZ - Arria V - Arria V GZ - Stratix[®] IV - Stratix V	- Control block-based device - Cyclone 10 GX - Arria 10 - SDM-based device - Stratix 10 - Agilex[™] 7 - Agilex 5 - Agilex 3

3. Features Support and Microarchitecture

All Nios V processor development tools and features in Quartus Prime Standard Edition software version 24.1 is similar to Quartus Prime Pro Edition version 25.3. However, there are some exceptions, as highlighted in the following table:

Table 2. Nios V Processor Core Features across Quartus Prime Software

Features	Quartus Prime Standard Edition Software	Quartus Prime Pro Edition Software
Development Tools	Does not support: - FreeRTOS-Plus TCP software package - Quartus Prime Embedded package - Nios V Signal Tap plugin	Supports: - FreeRTOS-Plus TCP software package - Quartus Prime Embedded package - Nios V Signal Tap plugin
All Nios V/c processor features	No exceptions	No exceptions
All Nios V/m processor features	Does not support: - ECC Correction Writeback - ECC Error Injection - ECC Error Exception	Supports: - ECC Correction Writeback - ECC Error Injection - ECC Error Exception
All Nios V/g processor features	Does not support: - Shadow Register in all firmware.⁽¹⁾ - CLIC interrupt mode.⁽¹⁾	Supports: - Shadow Register in Altera HAL and FreeRTOS firmware.⁽¹⁾ - CLIC interrupt mode in Altera HAL firmware only.⁽¹⁾

⁽¹⁾ Available operating systems in the Board Support Package Editor are Altera HAL, MicroC/OS-II, and FreeRTOS.

4. Board Support Package Editor

The following table shows the related steps that you can perform with the BSP Editor based on the applicable Quartus Prime software version.

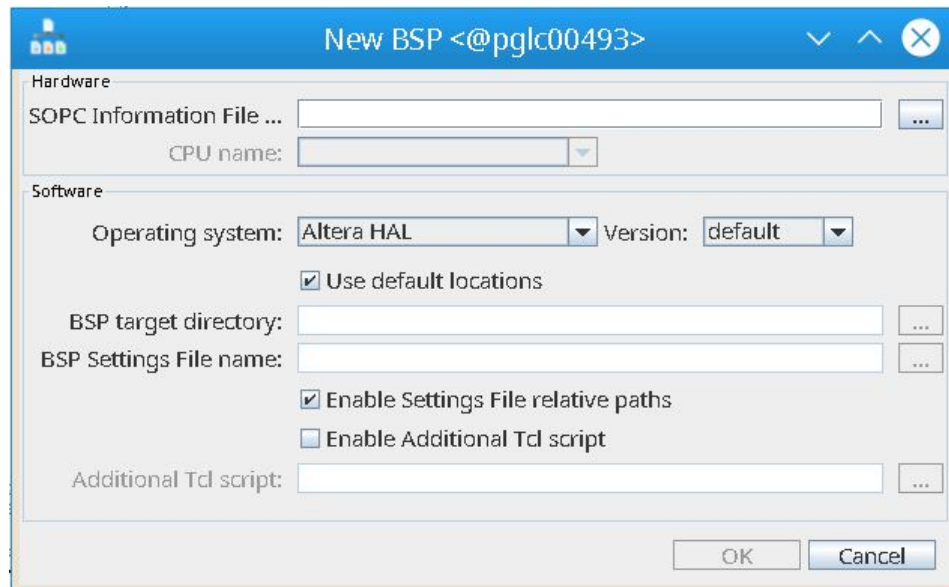
Table 3. Using the Board Support Package Editor in the Quartus Prime Software

Task	Quartus Prime Standard Edition Software	Quartus Prime Pro Edition Software
Invoking the BSP Editor	Run the <code>niosv-bsp-editor</code> command.	Use the Platform Designer.
Creating the BSP project	Provide the SOPCINFO file of the processor system into the BSP Editor.	Provide the QSYS file of the processor system into the BSP Editor.

4.1. Generating the BSP in Quartus Prime Standard Edition

1. Enter the Nios V Command Shell.
2. Invoke the BSP Editor with `niosv-bsp-editor` command.
3. In the BSP Editor, select **Main ► New BSP** to start your BSP project.

Figure 1. New BSP Window



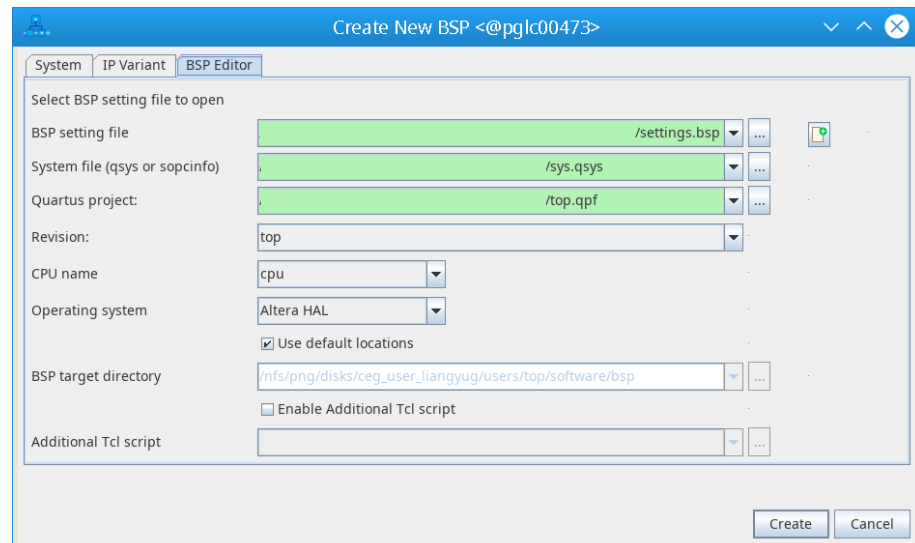
4. Configure the following settings:

- **SOPC Information File name:** Provide the SOPCINFO file (.sopcinfo).
 - **CPU name:** Select Nios V processor.
 - **Operating system:** Select the operating system of the Nios V processor.
 - **Version:** Leave as default.
 - **BSP target directory:** Select the directory path of the BSP project. You can pre-set it at <Project directory>/software/hal_bsp by enabling **Use default locations**.
 - **BSP Settings File name:** Type the name of the BSP Settings File.
 - **Additional Tcl script:** Provide a BSP Tcl script by enabling **Enable Additional Tcl script**.
5. Click **OK**, and a BSP project is started with a default BSP settings
 6. Configure the necessary BSP settings based on the application.
 7. Click **Generate BSP** to generate the BSP project.

4.2. Generating the BSP in Quartus Prime Pro Edition

1. Launch the Platform Designer.
2. In the Platform Designer window, select **File ► New BSP**

Figure 2. Create New BSP Window



3. Configure the following settings:
 - **BSP Settings File:** Provide the name of the new BSP Settings File.
 - **System file (qsys or socpinfo):** Provide the Nios V processor Platform Designer system file (.qsys).
 - **Quartus project:** Provide the Quartus Prime Project file (.qpf).
 - **Revision:** Select the correct project revision.
 - **CPU name:** Select the Nios V processor.

- **Operating system:** Select the operating system of the Nios V processor.
 - **BSP target directory:** Select the directory path of the BSP project. You can pre-set it at <Project directory>/software/hal_bsp by enabling **Use default locations**.
 - **Additional Tcl script:** Provide a BSP Tcl script by enabling **Enable Additional Tcl** script.
4. Click **OK**, and a BSP project is started with a default BSP settings
 5. Configure the necessary BSP settings based on the application.
 6. Click **Generate BSP** to generate the BSP project.

5. Document Revision History for AN 980: Nios V Processor Quartus Prime Software Support

Document Version	Changes
2025.11.21	Revised title for the topic <i>Features Support and Microarchitecture</i> and added the unsupported features to the <i>Nios V Processor Core Features in Different Quartus Prime Software</i> table.
2025.08.11	- Added <i>Supported features and microarchitecture</i> to the list of features in the topic: <i>Overview</i>. - Added <i>Agilex 3 for Quartus Prime Pro Edition Software</i> in the table <i>Nios V Processor Core and Device Support</i>. - Added topic <i>Supported Features and Microarchitecture</i>.
2024.04.01	Added <i>Agilex 5</i> in the table <i>Nios V Processor Core and Device Support</i> .
2023.12.11	- Updated Section: <i>Overview</i> to remove example design generation. - Updated Table: <i>Nios V Processor Core and Device Support</i>. - Removed Section: <i>Example Design Generation</i>.
2023.10.10	Corrected the steps in the topic <i>Generating the BSP in Quartus Prime Pro Edition</i> .
2023.10.02	Added <i>Nios V/c processor</i> in the table <i>Nios V Processor Core and Device Support</i> .
2023.05.26	Initial release.